
OPTICAL FORCES AND TORQUES ON PARTICLES

In 1986 Ashkin *et al.* (4.1, **Paper 4.1**) published their seminal paper on optical tweezers which describes the way in which a tightly focused laser beam may be used to trap and manipulate micron-sized particles in three dimensions. The paper has received over 400 scientific citations and, apart from its great importance in optical and atomic physics, has spawned a multitude of biological experiments including measurements of individual muscle forces, stretching of DNA and, when combined with other lasers, the cutting and manipulation of biological materials (4.2).

When placed in an electric field gradient, dielectric materials experience a force directed towards the region of highest field. The direction of the force does not depend upon the polarity of the field and so the AC field gradient associated with a tightly focused laser produces a force on a dielectric particle which moves it in the direction of the highest intensity of the beam. An additional force arises from light scattering which causes a re-direction of the input light's linear momentum into all directions. The goal in well-designed optical tweezers is to obtain a gradient force sufficiently large to overcome the scattering force and allow the formation of a 3D trap. The biggest contribution to the achievement of this goal comes from an extremely tight focusing of the trapping laser beam. Additional forces arise from the absorption of the light, which results in the transfer of both linear and angular momentum from the light beam to the particle. All of these forces have been observed in optical tweezers.

Ashkin showed by use of a ray-optical picture, that it was the refraction of the off-axis rays which were equivalent to the gradient force and that these could lift a transparent object towards the beam focus. The on-axis rays did not give rise to a gradient force, only a scattering force. He suggested that, in optical tweezers, an annular intensity profile should be preferable to an $HG_{0,0}$ mode (4.3) as this would result in a greater overall trapping force. Laguerre–Gaussian (LG) beams have been used for this purpose and a number of groups have demonstrated measurable improvements (4.4, 4.5). The on-axis intensity null associated with LG beams has also been used in optical tweezers to trap particles with a refractive index lower than that of the surrounding fluid (4.6). However, it should be emphasised that in all these applications the orbital angular momentum of the LG mode plays no role; it is the consequences of the annular intensity distribution which are being investigated.

The first use of a particle trapped in a Laguerre–Gaussian mode, and its associated orbital angular momentum, was reported in 1995 by He *et al.* (4.7, 4.8, **Paper 4.2**). They observed that when a tightly focused LG mode ($l = 3$) was directed at ceramic,

micron-sized particles, absorption of the light and its associated angular momentum set the particles into rotation about their axis. In this experiment, the particles were confined to the beam axis in two dimensions; the base of the sample cell provided containment in the third dimension. Subsequently, the same group showed that the additional spin angular momentum obtained by making the beam circularly polarised could cause the rotation of the particle to speed up or slow down (4.9, **Paper 4.3**).

In 1997, Simpson *et al.* (4.10, **Paper 4.4**) trapped a slightly absorbing Teflon particle at the focus of a circularly polarised LG mode. The circular polarisation corresponded to a spin angular momentum of $\hbar$ per photon and the $l = 1$ mode possessed an orbital angular momentum of $\hbar$ per photon. The particle was trapped in three dimensions in true optical tweezers and the absorption of the light gave a transfer mechanism for both the spin and orbital angular momentum from the beam to the particle. The insertion of a half waveplate into the beam changed the sign of the spin angular momentum, such that spin and orbital AM acted additively to give $2\hbar$ per photon or in opposite directions to give zero total angular momentum. The observed stop-start rotation of the particle confirmed the magnitude and sign of the orbital angular momentum term and, for $l = 1$, demonstrated the mechanical equivalence of spin and orbital angular momentum.

Optical tweezers have also been used to perform the microscopic equivalent of Beth's 1936 experiment (4.11, **Paper 1.2**), where the spin angular momentum associated with circular polarisation has been transferred to a birefringent element. Friese *et al.* (4.12) trapped microscopic fragments of calcite in circularly polarised optical tweezers. For calcite, the birefringence is high enough so that even microscopic particles change the polarisation state of the transmitted light. As with the Beth experiment, transformation of the light polarisation from circular to linear results in an angular momentum transfer of $\hbar$ per photon. They also demonstrated that a change to elliptically polarised light reduced the torque on the particle accordingly (4.13).

The interaction of orbital angular momentum with a cylindrical lens mode converter is the direct analogue to the transfer of spin angular momentum to a birefringent particle. To date no one has demonstrated the transfer of orbital angular momentum by the direct re-phasing of the component modes. However, light scattering offers a possible transfer mechanism for orbital angular momentum. While observing the behaviour of a non-absorbing metallic particle near the focus of a circularly polarised LG mode, O'Neil and Padgett (4.14) noted the particle rotated around the beam axis. It was found that the sense of circular polarisation made no difference to the motion of the particle so, clearly, spin angular momentum was not being transferred. In any case, such a transfer would have led to the particle being set into rotation about its own axis rather than that of the beam rather than about its own. They concluded that the scattering of light had caused the rotation. Even at the beam waist, the diameter of the LG mode exceeds that of a particle and the gradient force confines a dielectric particle to the annulus of maximum intensity which is at a uniform radius from the beam axis. If the particle were birefringent, then the change in the polarisation of the transmitted light would result in a subsequent transfer of spin angular momentum and the production of a torque acting on the particle about its own axis. If scattering is the predominant transfer mechanism, then the orbital angular momentum produces a torque on the particle about the beam axis. This spinning and orbiting of the particle is, therefore, identifiable with the spin and orbital angular momentum components respectively. A circularly polarised LG mode ($l = 8$) has been used to demonstrate that spin and orbital angular momentum may be observed to act independently within the same tweezers configuration (4.15, **Paper 2.7**).

REFERENCES

- 4.1 A Ashkin, JM Dziedzic, J E Bjorkholm and S Chu, 1986, *Opt. Lett.* **11** 288.
- 4.2 JE Molloy and MJ Padgett, 2002, *Contemp. Phys.* **43** 241.
- 4.3 A Ashkin, 1992, *Biophys. J.* **61** 569.
- 4.4 MEJ Friese, H Rubinsztein-Dunlop, NR Heckenberg and EW Dearden, 1998, *Appl. Opt.* **35** 7112.
- 4.5 AT O'Neil and MJ Padgett, 2001, *Opt. Commun.* **193** 45.
- 4.6 KT Gahagan and GA Swartzlander, 1996, *Opt. Lett.* **21** 827.
- 4.7 H He, NR Heckenberg and H Rubinsztein-Dunlop, 1995, *J. Mod. Opt.* **42** 217.
- 4.8 H He, MEJ Friese, NR Heckenberg and H Rubinsztein-Dunlop, 1995, *Phys. Rev. Lett.* **75** 826.
- 4.9 MEJ Friese, J Enger, H Rubinsztein-Dunlop and NR Heckenberg, 1996, *Phys. Rev. A* **54** 1593.
- 4.10 NB Simpson, K Dholakia, L Allen and MJ Padgett, 1997, *Opt. Lett.* **22** 52.
- 4.11 RA Beth, 1936, *Phys. Rev.* **50** 115.
- 4.12 MEJ Friese, TA Nieminen, NR Heckenberg and H Rubinsztein-Dunlop, 1998, *Nature* **394** 348.
- 4.13 MEJ Friese, TA Nieminen, NR Heckenberg and H Rubinsztein-Dunlop, 1998, *Opt. Lett.* **23** 1.
- 4.14 AT O'Neil and MJ Padgett, 2000, *Opt. Commun.* **185** 139.
- 4.15 AT O'Neil, I MacVicar, L Allen and MJ Padgett, 2001, *Phys. Rev. Lett.* **88** 053601.

Observation of a single-beam gradient force optical trap for dielectric particles

A. Ashkin, J. M. Dziedzic, J. E. Bjorkholm, and Steven Chu

AT&T Bell Laboratories, Holmdel, New Jersey 07733

Received December 23, 1985; accepted March 4, 1986

Optical trapping of dielectric particles by a single-beam gradient force trap was demonstrated for the first reported time. This confirms the concept of negative light pressure due to the gradient force. Trapping was observed over the entire range of particle size from 10 μm to ~ 25 nm in water. Use of the new trap extends the size range of macroscopic particles accessible to optical trapping and manipulation well into the Rayleigh size regime. Application of this trapping principle to atom trapping is considered.

We report the first experimental observation to our knowledge of a single-beam gradient force radiation-pressure particle trap.¹ With such traps dielectric particles in the size range from 10 μm down to ~ 25 nm were stably trapped in water solution. These results confirm the principles of the single-beam gradient force trap and in essence demonstrate the existence of negative radiation pressure, or a backward force component, that is due to an axial intensity gradient. They also open a new size regime to optical trapping encompassing macromolecules, colloids, small aerosols, and possibly biological particles. The results are of relevance to proposals for the trapping and cooling of atoms by resonance radiation pressure.

A wide variety of optical traps based on the basic scattering and gradient forces of radiation pressure have been demonstrated or proposed for the trapping of neutral dielectric particles and atoms.²⁻⁴ The scattering force is proportional to the optical intensity and points in the direction of the incident light. The gradient force is proportional to the gradient of intensity and points in the direction of the intensity gradient. The single-beam gradient force trap is conceptually and practically one of the simplest radiation-pressure traps. Although it was originally proposed as an atom trap,¹ we show that its uses also cover the full spectrum of Mie and Rayleigh particles.

It is distinguished by the feature that it is the only all-optical single-beam trap. It uses only a single strongly focused beam in which the axial gradient force is so large that it dominates the axial stability. In the only previous single-beam trap, the so-called optical levitation trap, the axial stability relies on the balance of the scattering force and gravity.⁵ In that trap the axial gradient force is small, and if one turns off or reverses the direction of gravity the particle is driven out of the trap by the axial scattering force.

There were also relevant experiments using gradient forces on Rayleigh particles that did not strictly involve traps, in which liquid suspensions of submicrometer particles acted as an artificial nonlinear optical Kerr medium.⁶

The physical origin of the backward gradient force in single-beam gradient force traps is most obvious for particles in the Mie size regime, where the diameter is large compared with λ . Here one can use simply ray optics to describe the scattering and optical momentum transfer to the particle.^{7,8} In Fig. 1a) we show the scattering of a typical pair of rays A of a highly focused beam incident upon a 10- μm lossless dielectric sphere, for example. The principal part of the momentum transfer from the incident light to the particle is due to the emergent rays A', which are refracted by the particle. Successive surface reflections, such as R_1 and R_2 , contribute a lesser scattering. For a glass particle in water the effective index m , equal to the index of the particle divided by the index of the medium, is about 1.1 to 1.2, and the sphere acts as a weak positive lens. If we consider the direction of the resulting forces F_A on the particle that are due to refraction of rays A in the weak-lens regime, we see as shown in Fig. 1a) that there is a substantial net backward trapping-force component toward the beam focus.

Figure 2 sketches the apparatus used for trapping Mie or Rayleigh particles. Spatially filtered argon-laser light at 514.5 nm is incident upon a high-numerical-aperture (N.A. 1.25) water-immersion microscope objective, which focuses a strongly convergent downward-directed beam into a water-filled glass cell. Glass Mie particles are introduced into the trap by an auxiliary vertically directed holding beam,⁵ which lifts particles off the bottom of the cell and manipulates them to the focus. Rayleigh particles are simply dispersed in water solution at reasonable concentrations and enter the trapping volume by Brownian diffusion. A microscope M is used to view the trapped particles visually off a beam splitter S or by recording the 90° scatter with a detector D.

Figure 1b) is a photograph of a 10- μm glass sphere of index about 1.6 trapped and levitated just below the beam focus of a ~ 100 -mW beam. The picture was taken through a green-blocking filter using the red fluorescence of the argon laser beam in water in order to make the trajectories of the incident and scattered

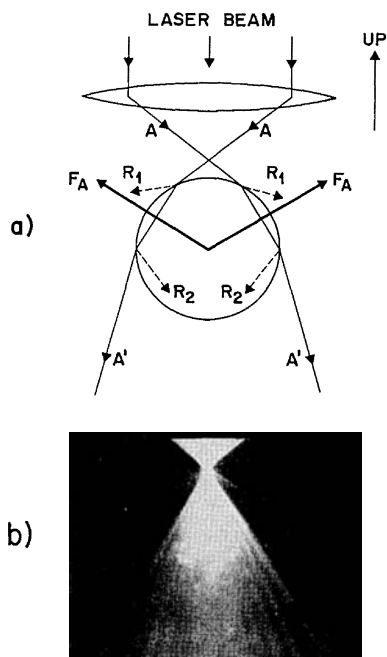


Fig. 1. a) Diagram showing the ray optics of a spherical Mie particle trapped in water by the highly convergent light of a single-beam gradient force trap. b) Photograph, taken in fluorescence, of a 10- μm sphere trapped in water, showing the paths of the incident and scattered light rays.

beams visible. The sizable decrease in beam angle of the scattered light, which gives rise to the backward force, is clearly seen. The stria in the forward-scattered light arise from the usual Mie-scattering ring pattern.

Next consider the possibility of single-beam trapping of submicrometer Rayleigh particles whose diameter $2r$ is much less than λ . Although we are now in the wave-optic regime, we will again see the role of the strong axial gradient in producing a net backward axial force component. For Rayleigh particles in a medium of index n_b the scattering force in the direction of the incident power is $F_{\text{scat}} = n_b P_{\text{scat}}/c$, where P_{scat} is the power scattered.⁹ In terms of the intensity I_0 and effective index m

$$F_{\text{scat}} = \frac{I_0}{c} \frac{128\pi^5 r^6}{3\lambda^4} \left(\frac{m^2 - 1}{m^2 + 2} \right)^2 n_b. \quad (1)$$

The gradient force F_{grad} in the direction of the intensity gradient for a spherical Rayleigh particle of polarizability α is⁶

$$F_{\text{grad}} = -\frac{n_b}{2} \alpha \nabla E^2 = -\frac{n_b^3 r^3}{2} \left(\frac{m^2 - 1}{m^2 + 2} \right) \nabla E^2. \quad (2)$$

This Rayleigh force component, in analogy with the gradient force for Mie particles, can be related to the lenslike properties of the scatterer.

As for atoms,¹ the criterion for axial stability of a

single-beam trap is that R , the ratio of the backward axial gradient force to the forward-scattering force, be greater than unity at the position of maximum axial intensity gradient. For a Gaussian beam of focal spot size w_0 this occurs at an axial position $z = \pi w_0^2/\sqrt{3} \lambda$, and we find that

$$R = \frac{F_{\text{grad}}}{F_{\text{scat}}} = \frac{3\sqrt{3}}{64\pi^5} \frac{n_b^2}{\left(\frac{m^2 - 1}{m^2 + 2} \right)} \frac{\lambda^5}{r^3 w_0^2} \geq 1, \quad (3)$$

where λ is the wavelength in the medium. This condition applies only in the Rayleigh regime where the particle diameter $2r \lesssim 0.2\lambda \approx 80 \text{ nm}$. In practice we require R to be larger than unity. For example, for polystyrene latex spheres in water with $m = 1.65/1.33 = 1.24$ and $2w_0 = 1.5\lambda = 0.58 \mu\text{m}$ we find for $R \geq 3$ that $2r \leq 95 \text{ nm}$. Thus with this choice of spot size we meet the stability criterion over the full Rayleigh regime. The fact that $R < 3$ for $2r > 95 \text{ nm}$ does not necessarily imply a lack of stability for such larger particles since we are beyond the range of validity of the formula. Indeed, as we enter the transition region to Mie scattering we expect the ray-optic forward-scattering picture to be increasingly valid. As will be seen experimentally we have stability from the Rayleigh regime, through the transition region, into the full Mie regime. For silica particles in water with $m = 1.46/1.33 = 1.10$ and $2w_0 = 0.58 \mu\text{m}$ we find for $R \geq 3$ that $2r \leq 126 \text{ nm}$. For high-index particles with $m \approx 3.0/1.33 = 2.3$ we find that $2r \leq 61 \text{ nm}$.

The stability condition on the dominance of the backward axial gradient force is independent of power and is therefore a necessary but not sufficient condition for Rayleigh trapping. As an additional sufficient trapping condition we have the requirement¹ that the Boltzmann factor $\exp(-U/kT) \ll 1$, where $U = n_b \alpha E^2/2$ is the potential of the gradient force. As was previously pointed out,⁶ this is equivalent to requiring that the time to pull a particle into the trap be much less than the time for the particle to diffuse out of the trap by Brownian motion. If we set $U/kT \geq 10$,

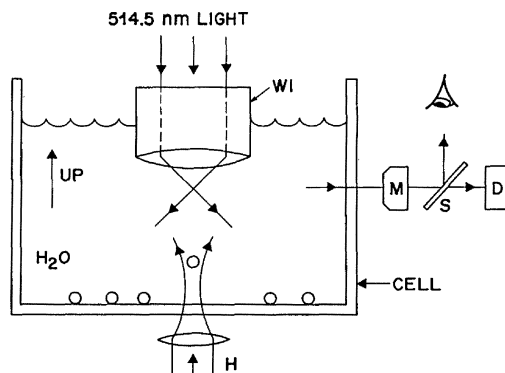


Fig. 2. Sketch of the basic apparatus used for the optical trapping of Mie and Rayleigh particles in water by means of a single-beam gradient force radiation-pressure trap.

for example, and use a power of ~ 1.5 W focused close to the limiting spot diameter of $0.58 \mu\text{m} \cong 1.5\lambda$, we find for silica that the minimum theoretical particle size that satisfies this condition is $2r = 19$ nm. For polystyrene latex, the minimum particle size that can be trapped under these conditions is $2r = 14$ nm. With a high-index particle of $m = 3/1.33 = 2.3$ the theoretical minimum size is $2r \cong 9$ nm.

Additional experiments were performed on individual colloidal polystyrene latex particles in water. Unfortunately the particles exhibit a form of optical damage at high optical intensities. For $1.0\text{-}\mu\text{m}$ spheres with a trapping power of a fraction of a milliwatt, particles survived for tens of minutes and then shrank in size and disappeared. Spheres of $0.173 \mu\text{m}$ were trapped for several minutes with a power of a milliwatt before being lost. Particles of $0.109\text{-}\mu\text{m}$ diameter required about 12–15 mW and survived about 25 sec. With 85- and 38-nm latex particles the damage was so rapid that it was difficult to observe the scattering reliably. It was nevertheless clear that trapping occurred over full size range from Mie to Rayleigh particles.

The remarkable uniformity of latex particles was evident from the small variation of $\pm 15\%$ in the 90° scatter of $0.109\text{-}\mu\text{m}$ particles. Since the scattering is closely Rayleigh this corresponds to a diameter variation of $\pm 2.4\%$. Subsequently we determined the size of unknown silica Rayleigh particles by comparing their scatter with the scatter from the $0.109\text{-}\mu\text{m}$ particles taken as a standard, using Eq. (1). Although the $0.109\text{-}\mu\text{m}$ particles are not strictly Rayleigh, one can make a modest theoretical correction¹⁰ of ~ 1.06 to the effective particle size.

Trapping of nominally spherical colloidal silica particles was observed by using commercially available Nalco and Ludox samples¹¹ diluted with distilled water. With a high concentration we quickly collect many particles in the trap and observe a correspondingly large scattering. At reduced concentration we can observe single particles trapped for extended time. Once a particle is captured at the beam focus we observe an apparent cessation of all Brownian motion and a large increase in particle scattering.

With silica samples we always observe a wide distribution of particle sizes as evidenced by the more than an order-of-magnitude difference in scattering from particles trapped with a given laser power. Particle damage by the light was not a serious problem with silica particles. The smaller particles of the distribution showed only slight changes of scattering over times of minutes. The larger particles would often decay by factors up to 3 in comparable times.

Measurements were made on a Nalco 1060 sample with a nominal size of about 60 nm and an initial concentration of silica of 50% by weight diluted to one part in 10^5 – 10^6 by volume. Trapping powers of 100–400 mW were used. The absolute size of the Nalco 1060 particles as determined by comparison with the $0.109\text{-}\mu\text{m}$ latex standard varied from ~ 50 to

90 nm with many at ~ 75 nm. We also studied smaller silica particles using a Ludox TM sample with nominal particle diameter of ~ 21 nm and a Nalco 1030 sample with nominal distribution of 11 to 16 nm. Dilutions of $\sim 10^6$ – 10^7 and powers of ~ 500 mW to 1.4 W were used. With both samples we were limited by laser power in the minimum-size particle that could be trapped. With 1.4 W of power the smallest particle trapped had a scattering that was a factor of $\sim 3 \times 10^4$ less than from the $0.109\text{-}\mu\text{m}$ standard. This gives a minimum particle size of 26 nm assuming a single spherical scatterer. The measured minimum size of 26 nm compares with the theoretically estimated minimum size of ~ 19 nm for this power, based on $U/kT = 10$ and spot size $2w_0 = 0.58 \mu\text{m}$. This difference could be resolved by assuming a spot size $2w_0 = 0.74 \mu\text{m} = 1.28 (0.58 \mu\text{m}) \cong 1.9 \lambda$.

Experimentally we found that we could introduce a significant drift of the fluid relative to the trapped particle by moving the entire cell transversely relative to the fixed microscope objective. This technique gives a direct method of measuring the maximum trapping force. It also implies the ability to separate a single trapped particle from surrounding untrapped particles by a simple flushing technique.

Our observation of trapping of a 26-nm silica particle with 1.4 W implies, by simple scaling, the ability to trap a 19.5-nm particle with $m = 1.6/1.33 = 1.20$ and a 12.5-nm particle of $m = 3.0/1.33 = 2.26$ at the same power. These results suggest the use of the single-beam gradient force traps for other colloidal systems, macromolecules, polymers, and biological particles such as viruses. In addition to lossless particles with real m there is the possibility of trapping Rayleigh particles with complex m for which one can in principle achieve resonantly large values of the polarizability α . Finally, we expect that these single-beam traps will work for trapping atoms¹ as well as for macroscopic Rayleigh particles since atoms can be viewed as Rayleigh particles with a different polarizability.

References

1. A. Ashkin, Phys. Rev. Lett. **40**, 729 (1978).
2. A. Ashkin, Science **210**, 1081 (1980); V. S. Letokhov and V. G. Minogin, Phys. Rep. **73**, 1 (1981).
3. A. Ashkin and J. P. Gordon, Opt. Lett. **8**, 511 (1983).
4. A. Ashkin and J. M. Dziedzic, Phys. Rev. Lett. **54**, 1245 (1985).
5. A. Ashkin and J. M. Dziedzic, Appl. Phys. Lett. **19**, 283 (1971).
6. P. W. Smith, A. Ashkin, and W. J. Tomlinson, Opt. Lett. **6**, 284 (1981); and A. Ashkin, J. M. Dziedzic, and P. W. Smith, Opt. Lett. **7**, 276 (1982).
7. A. Ashkin, Phys. Rev. Lett. **24**, 146 (1970).
8. G. Roosen, Can. J. Phys. **57**, 1260 (1979).
9. See, for example, M. Kerker, *The Scattering of Light* (Academic, New York, 1969), p. 37.
10. W. Heller, J. Chem. Phys. **42**, 1609 (1965).
11. Nalco Chemical Company, Chicago, Illinois; Ludox colloidal silica by DuPont Corporation, Wilmington, Delaware.

Direct Observation of Transfer of Angular Momentum to Absorptive Particles from a Laser Beam with a Phase Singularity

H. He, M. E. J. Friese, N. R. Heckenberg, and H. Rubinsztein-Dunlop

Department of Physics, The University of Queensland, Brisbane, Queensland, Australia Q4072

(Received 28 November 1994; revised manuscript received 4 April 1995)

Black or reflective particles can be trapped in the dark central minimum of a doughnut laser beam produced using a high efficiency computer generated hologram. Such beams carry angular momentum due to the helical wave-front structure associated with the central phase singularity even when linearly polarized. Trapped absorptive particles spin due to absorption of this angular momentum transferred from the singularity beam. The direction of spin can be reversed by changing the sign of the singularity.

PACS numbers: 42.25.Md, 42.40.My

It is well known that a circularly polarized beam carries angular momentum. Each photon of such a beam has an angular momentum of $\hbar$. The effects produced by the optical angular momentum are hard to observe in most circumstances as they represent extremely small quantities. The angular momentum flux carried by a circularly polarized 10 mW He-Ne laser beam is of the order of 10^{-18} mN. The first attempt to measure the torque produced by the optical angular momentum was made by Beth [1] 59 years ago. Beth reported that his results agreed with theory in sign and magnitude. More recently Santamato *et al.* [2] observed the light induced rotation of liquid crystal molecules, Chang and Lee [3] calculated the optical torque acting on a weakly absorbing optically levitated sphere, and Ashkin and Dziedzic [4] observed rotation of particles optically levitated in air.

A linearly polarized wave containing a central phase singularity also carries angular momentum associated with its helical structure. Figure 1 shows this structure for a TEM_{01}^* beam [5]. Each photon carries $l\hbar$ angular momentum where l is called the topological charge of the singularity. This contribution to the angular momentum is sometimes referred to as "orbital angular momentum" to

distinguish it from "spin" angular momentum associated with circular polarization [6,7].

Allen *et al.* [6] have proposed to measure the angular momentum carried by a doughnut laser beam by measuring the torque acting on an optical device which reverses the chirality of the beam.

We have demonstrated in a previous paper [8] that black or reflective particles of sizes of 1–2 μm can be trapped optically in a liquid by higher order doughnuts produced by high efficiency computer generated holograms. We also mentioned that some bigger absorptive particles were set into rotation by slightly defocused doughnuts. In this paper we report on further experiments concerned with the transfer of angular momentum to absorptive particles and their subsequent rotation. The detection is performed using a video camera. A large number of small particles have been trapped and have been recorded. Our results clearly show that the rotation directions of all these trapped particles agree with the sign of the doughnut.

In this Letter we analyze the results of the rotating particles in terms of possible torques acting on a trapped absorptive particle.

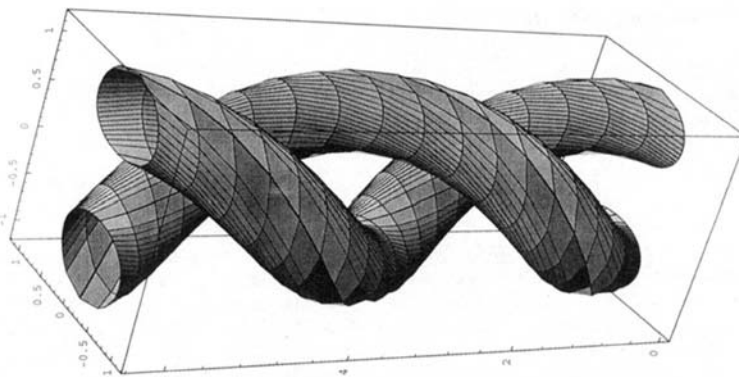


FIG. 1. Snapshot of the irradiance structure of a TEM_{01}^* $[(r/\omega)e^{-r^2/\omega^2}e^{i\theta}e^{ikz}]$ beam containing a first order phase singularity. The surface represented is that where the irradiance has half its peak value and we take $k = 1$, $\omega = 1$ for simplicity.

The angular momentum carried by light can be understood in two ways. Classically, electromagnetic radiation carries momentum, which can be both linear and angular. Allen *et al.* [6] have shown that for a Laguerre-Gaussian beam the angular momentum density is given as

$$M_z = \frac{l}{\omega} |u|^2 + \frac{\delta_z r}{2\omega} \frac{\partial |u|^2}{\partial r}, \quad (1)$$

where u is the amplitude of the light field, l is the azimuthal index number of the Laguerre polynomial, $\delta_z = \mp 1$ for right-handed or left-handed circularly polarized light and $\delta_z = 0$ for linearly polarized light.

Quantum mechanically, one would say that each photon carries $(l \mp \delta_z)\hbar$ of angular momentum because of the well-known analogy between paraxial theory and quantum mechanics [9].

Although the above simple relationship is valid only within the paraxial approximation as shown recently by Barnett and Allen [7], for a linearly polarized light ($\delta_z = 0$), even when tightly focused as in an optical trapping experiment, the total angular momentum per second is still given by

$$\Gamma_z = \frac{P}{\omega} l, \quad (2)$$

where P is the laser power.

A linearly polarized charge 3 singularity beam corresponds to $l = 3$, $\delta_z = 0$.

This suggests that an absorptive particle illuminated by such a singularity beam should be set into rotation in the same sense as the helical beam. It should reverse its rotation direction when the direction of rotation of the helical wave is reversed.

The experimental setup for the optical trap used in this work is similar to the one used in [10]. A linearly polarized TEM₀₀ laser beam from a 15 mW He-Ne laser (632.8 nm) illuminates a blazed charge 3 phase hologram. The hologram then produces a phase singular laser doughnut beam equivalent to a linearly polarized Gauss-Laguerre LG₀₃ mode with a power of approximately 7 mW. As the hologram used here is blazed, the sign of the doughnut can be simply reversed by turning the hologram around. We can also switch the diffracted beam between doughnut mode and Gaussian mode by moving the hologram sideways. The laser beam is now introduced into a microscope (Olympus CHT) through the aperture for the vertical illuminator. A dichroic mirror reflects the beam to fill the back aperture of an oil-immersion, high numerical aperture ($NA = 1.30$), $100\times$ objective. The tightly focused doughnut beam had a diffraction limited beam waist within micrometers of the object plane of the objective.

The particles were illuminated from below by a lamp and green filters. A video camera placed vertically on the top of the microscope was used to record the motion of the trapped particles. We normally used black high-

T_c superconductor ceramic powder as absorptive particles, dispersed in kerosene. Sizes of the particles that can be trapped are around $1\text{--}2\ \mu\text{m}$. We have performed similar experiments using CuO particles dispersed in water.

Using the charge 3 doughnut, we can trap a particle in the dark central spot. It can then be moved around relative to its surroundings by moving the microscope slide. The maximum measured trapping speed is around $5\ \mu\text{m/s}$. Particles adhering to the slide can often be set free by switching from the doughnut to the Gaussian mode and "kicking" them with the strong repulsive force.

Using a video camera, we clearly see a trapped particle rotating always in one direction determined by the helicity of the beam. The particle also keeps rotating in the same direction while being moved relative to its surroundings. The rotation of a trapped particle can be easily maintained for long periods of time. The particle can be set free and trapped again with the same rotation direction. In a session, more than 30 particles have been trapped, moved, set free, and trapped again. They all rotate in the same direction as the helicity of the beam. A few particles with very irregular shapes tumble wildly and occasionally rotate in the opposite direction for one or two turns but much more slowly than they rotate the other way. However, most of the time they rotate in the same direction as other particles. Those particles with close to spherical shapes always rotate in the same direction with high constant speed.

Turning the hologram back to front, that is reversing the charge of the singularity, we repeat the above procedures and we observe the same effects except that the rotation direction of all particles is reversed.

The trapped particle rotation speed varies from 1 to 10 Hz depending on shape and size.

Figure 2 shows six successive frames of a video recording. A fairly asymmetric particle, about $2\ \mu\text{m}$ across, tumbling and rotating at about 1 Hz, has been chosen for illustration. It lies near the top of the frame and over a period of 0.24 s rotates through an angle of a little over $\pi/2$. Surrounding objects are stationary. More rapidly rotating symmetrical particles show up poorly in individual frames. However, the rotation is clear in continuous playback, in spite of motional blurring, very limited depth of field, and the fact that the microscope is operating near its limit of resolution.

When trapping CuO particles in water, if a little detergent is added, all the particles move very freely, and when one is trapped, almost immediately surrounding particles begin to move in radially and join a general circulation about it, in the same direction as the rotation of the trapped one.

Taking losses into consideration, the power incident on the focal plane is around 4 mW. According to Eq. (2), the angular momentum flux of such a doughnut beam produced by a charge 3 blazed hologram illuminated by a 633 nm HeNe laser is about 4×10^{-18} mN.

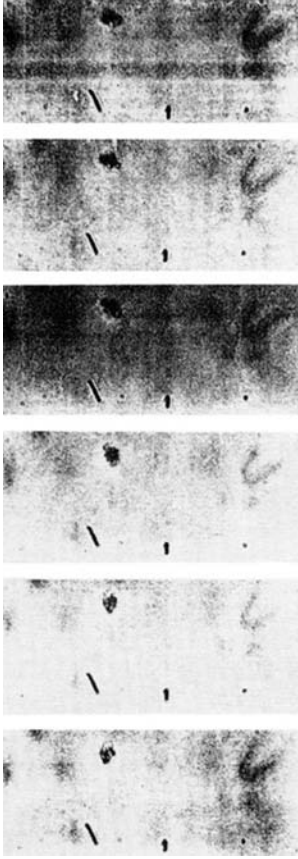


FIG. 2. Six successive frames of a video recording of a particle of black ceramic trapped in a charge three doughnut beam. The particle is near the top; other objects in the field are stationary. A rotation of just over $\pi/2$ occurs in the period shown.

Assume a particle absorbs 25% of the beam and hence 25% of the angular momentum. The rotation speed will become constant when the torque produced by the doughnut laser beam is balanced by the drag torque exerted by the surrounding liquid.

The drag torque acting on a spherical particle rotating with angular velocity ω_a is given by [11]

$$\tau_v = -8\pi\eta r^3\omega_a, \quad (3)$$

where r is the radius of the particle, η is the viscosity of the liquid, taken as $1.58 \times 10^{-3} \text{ N s m}^{-2}$ for kerosene.

For our particles, with radius about $1 \mu\text{m}$, this leads to a rotation speed of around 4 Hz, consistent with our observations, bearing in mind the actual form of our particles, and the fact that we have neglected the effects of the nearby slide surface. Our assumption that the particles absorb 25% of the incident power is based on the case where the sizes of particles and the central dark spot are

the same. We define the size of the dark spot to be the diameter of the maximum intensity ring of the doughnut beam. This estimate does not affect the relative sizes of the torques hypothetically present as they all would be proportional to the absorbed power.

Although absorption of angular momentum therefore accounts satisfactorily for the observed rotation, we briefly consider other explanations that might be advanced. It is possible that an unbalanced force, such as a thermal force, or a scattering force, may produce a torque acting upon a nonsymmetric particle. Among these forces, the scattering force is the biggest. However, since the direction of the scattering force is predominantly downward, the direction of the torque produced by the scattering force is therefore perpendicular to the beam axis. We believe that such torques are responsible for the irregular tumbling exhibited by asymmetric particles.

However, it is conceivable that reflections on the surface of an asymmetric particle may produce a torque in the same direction as that of the angular momentum from the beam.

Such a torque will be, at most,

$$\tau_a = k\gamma Fr, \quad (4)$$

where F is the force acting on the particle, r is an effective radius, γ is the reflection coefficient, and k is an asymmetry index, expressing the ratio of the area of a typical irregularity to the total area.

The scattering force (radiation pressure) cannot exceed

$$F_s = \frac{P}{c} = 1.3 \times 10^{-11} \text{ N}.$$

The value of k can vary from 0 to 1. However, only if the particle has a perfect “propeller” shape will k have a value approaching 1. Since our particles are mostly spheroidal and the effect of the trap is to center them in the beam, as the main asymmetry is due to the bumps on the surface of the particle, the value of k is approximately equal to the ratio of the bump area to the particle cross sectional area. Using a scanning electron microscopy image of these particles, we estimate that the value of k will normally be less than 1×10^{-2} as the irregularities are smaller than 10%. It is safe to assume the reflection coefficient to be smaller than 0.1 as our particles are highly absorptive. Hence, the torque due to asymmetric scattering forces is less than 10^{-20} N m , which is much smaller than the angular momentum from the doughnut laser beam. Obviously, a particle with an irregular shape may have a large value of k for a short period of time during which the scattering torque may dominate. However, the scattering torque is not constant, and hence the rotation direction changes. Furthermore, such a scattering force would be independent of the sign of the doughnut and therefore not able to explain our experimental results.

It is very difficult to estimate the magnitude of possible thermal effects like the photophoretic forces known to

act on illuminated particles in air [12], but they could be expected to be smaller in our experiments because of the much higher thermal conductivities of the liquid media and the particles used. We have been unable to find any reports of such forces in liquids. Even if such forces are acting, considering the high thermal conductivity of the material and the fact that irregularities are small relative to the particle size and the wavelength, it is unlikely that they would exert torques leading to regular rotation with direction depending on the helicity of the wave.

Finally, if the rotation we observe was due to some thermally mediated torque, in order that angular momentum be conserved, it would be necessary that the surrounding liquid circulate in the opposite direction to the particle. However, our observations of the motion of nearby particles clearly eliminates this possibility. We see nearby particles swept around in the same direction as the rotating particle, consistent with a picture of an externally driven trapped particle stirring the surrounding liquid.

In conclusion, it has been demonstrated that absorptive particles trapped in the dark central minimum of a doughnut laser beam are set into rotation. The rotational motion of the particles is caused by the transfer of angular momentum carried by the photons. Since the laser beam is linearly polarized, this must originate in the "orbital" angular momentum associated with the helical wave-front structure and central phase singularity. We have shown that the direction of the rotational motion is determined by the chirality of the helical wave front. With a laser power of a few milliwatts, the rotation speed of the particles lies

between 1 and 10 Hz depending on their sizes and shapes. This is in agreement with a simple model of absorption of the angular momentum of the radiation field.

We would like to thank Professor P. Drummond for useful discussions. We also want to thank R. McDuff and A. Noskoff for assistance with the images. This work was supported by the Australian Research Council.

-
- [1] R. A. Beth, Phys. Rev. **50**, 115 (1936).
 - [2] E. Santamato, B. Daino, M. Romagnoli, M. Settembre, and Y. R. Shen, Phys. Rev. Lett. **57**, 2433 (1986).
 - [3] S. Chang and S. S. Lee, J. Opt. Soc. Am. B **2**, 1853 (1985).
 - [4] A. Ashkin, Science **210**, 4474 (1980); **210**, 1081 (1980).
 - [5] N. R. Heckenberg, R. G. McDuff, C. P. Smith, H. Rubinsztein-Dunlop, and M. J. Wegner, Opt. Quant. Electron. **24**, S951 (1992).
 - [6] L. Allen, M. W. Beijersbergen, R. J. C. Spreeuw, and J. P. Woerdman, Phys. Rev. A **45**, 8185 (1992).
 - [7] S. M. Barnett and L. Allen, Opt. Commun. **110**, 670 (1994).
 - [8] H. He, N. R. Heckenberg, and H. Rubinsztein-Dunlop, J. Mod. Opt. **42**, 217 (1995).
 - [9] D. Marcuse, *Light Transmission Optics* (Van Nostrand, New York, 1972).
 - [10] C. D'Heron, E. W. Dearden, H. Rubinsztein-Dunlop, and N. R. Heckenberg, J. Mod. Opt. **41**, 595 (1994).
 - [11] S. Oka, in *Rheology*, edited by F. R. Eirich (Academic Press, New York, 1960), Vol. 3.
 - [12] L. R. Eaton and S. L. Niste, AIAA J. **17**, 261 (1979).

Optical angular-momentum transfer to trapped absorbing particles

M. E. J. Friese,¹ J. Enger,² H. Rubinsztein-Dunlop,¹ and N. R. Heckenberg¹

¹Department of Physics, The University of Queensland, Queensland, Australia 4072

²Department of Physics, Chalmers University of Technology, S-41296 Göteborg, Sweden
(Received 21 November 1995)

Particle rotation resulting from the absorption of light carrying angular momentum has been measured. When absorbing CuO particles ($1 - 5 \mu\text{m}$) were trapped in a focused “donut” laser beam, they rotated, due to the helical phase structure of the beam. Changing the polarization of the light from plane to circular caused the rotation frequency to increase or decrease, depending on the sense of the polarization with respect to the helicity of the beam. Rotation frequencies were obtained by Fourier analysis of amplitude fluctuations in the backscattered light from the particles. [S1050-2947(96)08908-1]

PACS number(s): 42.50.Vk, 42.25.Ja, 42.40.My

In this article, we report on the transfer of both orbital and spin angular momentum from a polarized Laguerre-Gaussian (LG) mode laser beam to macroscopic particles. The angular momentum carried by light can be characterized by the “spin” angular momentum associated with circular polarization [1] and the “orbital” angular momentum associated with the spatial distribution of the wave [2].

To our knowledge, first experimental observation of the torque on a macroscopic object resulting from interaction with light was by Beth [1] in 1936, who observed the deflection of a quartz wave plate suspended from a thin quartz fiber when circularly polarized light passed through it. An experiment was proposed in 1957 [3] to measure radiation torque using microwave radiation, and measurements of light torques were made in 1966 [4]. The experiments of Allen [4] showed that the torque on a suspended dipole due to circularly polarized radiation increased linearly with the intensity of the light. Recently, Allen *et al.* [2] have shown that a Laguerre-Gaussian laser mode has a well defined orbital angular momentum, and proposed an experiment analogous to the experiment of Beth, to observe the torque on suspended cylindrical lenses arising from the reversal of the helicity of a LG mode. In 1995, He *et al.* [5] observed transfer of angular momentum from a linearly polarized LG mode laser beam to absorbing particles. Micrometer-sized CuO particles were trapped in a focused “donut” beam, and observed to rotate when viewed through an optical microscope. The direction of the particle rotation was found to be determined by the direction of the helicity of the beam.

We present measurements of rotation frequency of micron-sized black CuO particles trapped in a donut laser beam, for different states of polarization of the incident beam. The measurements clearly show that an absorbing particle trapped and rotating in a focused plane-polarized donut laser beam rotates faster if the beam is changed to circularly polarized with spin of the same sense as the helicity of the donut, and slower if the beam is changed to circularly polarized with spin of the opposite sense to that of the helicity. The magnitude of the change in rotation frequency agrees with that expected from theory [6], within the accuracy of the experiment.

The transfer of angular momentum from light to an absorbing particle can be understood using classical electro-

magnetic theory. The torque due to the polarization of the light on a particle that absorbs power P_{abs} is $\Gamma_{\sigma_z} = P_{abs} \sigma_z / \omega$, where σ_z is ± 1 for circularly polarized light and 0 for plane-polarized light and ω is the frequency of the light. Our experiments were conducted using what is essentially a LG mode, which has an angular dependence of $e^{-il\phi}$, where l is the azimuthal mode index, or charge [7]. Such beams have a donut shape with a central zero. Even when plane polarized, the field has a helical structure due to the $e^{-il\phi}$ term. The torque on an absorbing particle, which absorbs power P_{abs} , from a plane-polarized LG mode beam, determined using electromagnetic theory [2], is $\Gamma_l = P_{abs} l / \omega$. In the paraxial limit, the torque on the particle due to both the polarization and the helical Poynting vector of the LG mode is simply [2]

$$\Gamma = \frac{P_{abs}}{\omega} (l + \sigma_z). \quad (1)$$

More intuitively, the light torque can be seen to arise from the angular momentum of photons. Each photon of energy $\hbar\omega$ can be assigned a “spin” of $\sigma_z \hbar$. The LG modes can be seen as eigenmodes of the angular-momentum operator L_z [2] and thus carry an “orbital” angular momentum of $l\hbar$ per photon. The angular momentum carried by a photon of a polarized Laguerre-Gaussian mode is then $(l + \sigma_z)\hbar$. When this is multiplied by the number of photons absorbed per second, the same resulting torque is obtained as that found using electromagnetic theory.

However, this very simple result is restricted to the paraxial approximation. Barnett and Allen [6] have developed a general nonparaxial theory, using a mode which is a general LG mode in the paraxial limit. The simple dichotomy between orbital and spin angular momentum that exists in the paraxial limit is no longer found. Instead, the results of Barnett and Allen imply that the torque will be

$$\Gamma = \frac{P_{abs}}{\omega} \left\{ (l + \sigma_z) + \sigma_z \left(\frac{2kz_R}{2p + l + 1} + 1 \right)^{-1} \right\}. \quad (2)$$

Here p and l are the mode indices, k is the wave number, and z_R is a length term, which in the paraxial limit is associated with the Rayleigh range [6]. As the beam becomes

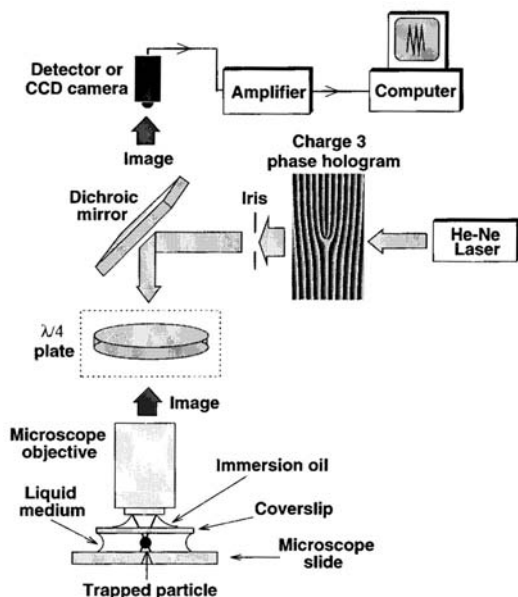


FIG. 1. Experimental setup for observation of optical angular-momentum transfer.

more strongly focused, the mixed term in Eq. (2) becomes more significant. However, if the laser beam were focused to one wavelength across, the most this term could contribute would be 20%, and substituting values appropriate to our experiments, the contribution for our laser beam (which closely approximates a Laguerre-Gaussian LG_{03} mode in the far field) focused to a $2\text{-}\mu\text{m}$ waist is only 4%. Thus we expect to observe the torques on a particle trapped in a LG_{03} mode laser beam of left-circular, plane, and right-circular polarization to be approximately in the ratio 2:3:4.

The quantity P_{abs} can, in principle, be determined from the surface area and absorptivity α of the particles and the beam profile [5]. A body rotating in a viscous fluid also experiences a viscous drag torque which is proportional to the angular velocity $\omega_{particle}$. An equilibrium angular velocity will be reached when the two torques balance, so the rotation angular frequency can be used as a measure of the optical torque.

The essential result to be tested in our experiments is thus that the angular momentum carried by photons of a left-circularly polarized, plane polarized, and right-circularly polarized charge-3 donut beam is in the ratio 2:3:4 (for a right-helical beam). This translates to the angular velocity of a particle trapped in such a beam changing in the same 2:3:4 ratio as the polarization is changed.

Our experiments were performed using an *optical tweezers* setup, as shown in Fig. 1. A Gaussian (TEM_{00}) beam from a 17-mW He-Ne laser was passed through a computer-generated phase hologram [8], which produced a beam closely approximating a Laguerre-Gaussian LG_{03} mode in the far field [7]. This donut beam was then introduced into an optical microscope. A quarter-wave plate was introduced into the beam path directly before a $100\times$ microscope ob-

jective (N.A. 1.3), which focused the donut beam to a waist approximately $2\text{ }\mu\text{m}$ in diameter.

As explained above, the objective of this experiment was to determine if transfer of spin angular momentum from light to absorbing particles is observable on a macroscopic scale. The $\lambda/4$ plate, (as shown in Fig. 1) was used to change the polarization of the donut beam. Since the reflectivity of the beamsplitter used to direct the laser beam into the microscope objective is polarization dependent, the $\lambda/4$ plate was placed directly before the objective, after the beamsplitter, in the beam path. The polarization was checked after the objective lens using an analyzer, and the power of the beam was measured for each polarization (left- and right-circularly polarized and plane polarized.) The laser power varied less than 1% between different polarizations. The $\lambda/4$ plate was also carefully positioned so that the beam alignment into the objective did not change when varying the state of polarization of the incoming beam, since poor alignment can cause unstable trapping and thus affect the rotation speed of trapped particles. The samples of absorbing particles (CuO) of sizes up to $20\text{ }\mu\text{m}$, in kerosene mixed with oil to optimize the stability of the trap) were placed between a glass microscope slide and coverslip.

Absorbing particles of size $1\text{--}5\text{ }\mu\text{m}$ can be trapped above the focus of the laser beam, where the spot size is rapidly changing. This can be easily understood by considering that Poynting vector and hence the force due to radiation pressure has a component toward the center of the beam as well as one in the direction of propagation. Thus absorbing particles in a converging donut beam should experience a force trapping them in the center and pushing them in the direction of propagation, so they are pressed to the glass surface and trapped in the transverse plane.

The helical structure of the beam means that the Poynting vector also has a tangential component that causes the CuO particles to begin to rotate as soon as they are trapped [5]. The rotation frequency of particles trapped in a linearly polarized helical donut beam has previously been measured by viewing video recordings of rotating particles frame by frame. However, this method is difficult and of limited accuracy as the particles move out of focus when they are trapped and thus their images are not well defined on video. In the present experiments, the rotation frequency was measured using a photodetector positioned off center to intercept a portion of the light reflected from the rotating particle. The particles are irregularly shaped, and protruding parts of the particle reflect a "flash" of light onto the photodetector. The signal detected by the photodetector is then processed using Fourier transform methods, and the transform yields the rotation frequency of the particle. If the particle has two protruding parts positioned approximately opposite one another, a peak at double the rotation frequency shows up in the spectrum as well as the peak due to the actual rotation frequency. An example of such a signal is shown in Fig. 2, and the resulting spectrum is shown in Fig. 3. That the main peak shown in Fig. 3 corresponds to the rotation frequency of the particle was verified by a frame-by-frame study of a simultaneous video recording of the same particle rotating. However, the sensitivity of the reflected signal method is such that measurements may be made on less asymmetric particles than can be easily followed in video recordings. This

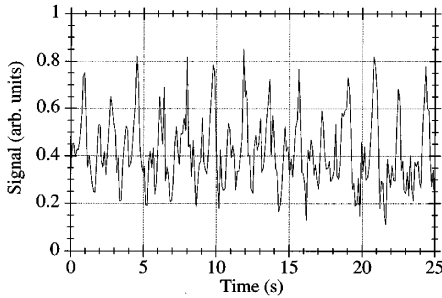


FIG. 2. Sample of raw signal detected from a rotating CuO particle. This particle has two comparatively large protrusions approximately opposite each other.

results in smoother rotation, monitored over longer periods, with a consequent great improvement in the accuracy of the rotational velocity estimates.

The sequence of the experiment was as follows: CuO particles were trapped using a plane-polarized donut beam. The photodetector was used to measure the reflected light, and the signal was sampled at 20 Hz for a period of 100 sec. The $\lambda/4$ plate was then rotated to give left-circularly polarized light while the particle remained trapped, and the signal was again sampled. This procedure was repeated for right-circularly-polarized light and again for plane-polarized light, in that order. In this way, the rotation frequency for a particular particle was obtained for each polarization. Variations of the order in which the polarization was changed were also used. For each experiment, the first and final data sets were taken with the laser beam in the same polarization state, to ensure that any systematic increase or decrease in rotation speed would be evident. The spectra obtained (Fig. 4) clearly show an increase or decrease in rotation speed of the particle due to the circular polarization of the light. The rotation frequency increases when the helicity of the electric field vector (due to the circular polarization) has the same direction as

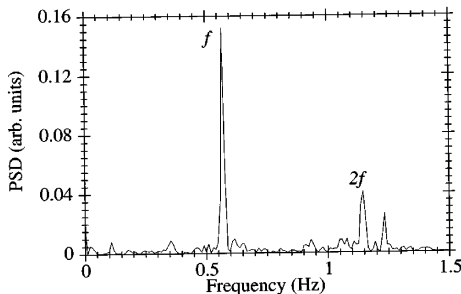


FIG. 3. Spectrum obtained from a CuO particle with two protrusions resulting in strong peaks at both f and $2f$, where f is the actual rotation frequency of the particle. That f is indeed the true rotation frequency of the particle was verified by observation of a video recording made simultaneously with the taking of the data shown in Fig. 2.

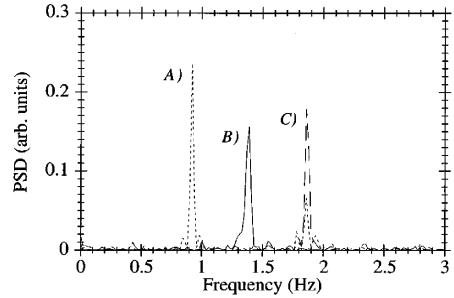


FIG. 4. Typical spectra showing the change in rotation frequency of a trapped particle as the polarization of the beam is changed. (A) is the peak corresponding to rotation in a left-circularly-polarized right-helical LG_{03} beam, (B) corresponds to a plane-polarized right-helical LG_{03} beam, and (C) corresponds to rotation in a right-circularly-polarized right-helical LG_{03} beam.

the helicity of the Poynting vector (due to the donut mode), and the decrease in rotation frequency corresponds to the case where the torque due to polarization opposes the torque due to the Laguerre-Gaussian mode. To verify this observation, the hologram which is used to produce the Laguerre-Gaussian mode was turned around, reversing the sense of the helicity of the Poynting vector. The trapped particles now all rotate in the opposite direction to the previous situation, and are slowed down by the polarization which previously caused an increase in rotation frequency and sped up by the polarization which previously slowed them down. Spectra obtained for this orientation of the donut are of the same quality as that shown in Fig. 4.

We have plotted the fractional increase or decrease in rotation frequency due to circular polarization against the rotation frequency for plane-polarized light in Fig. 5. From this graph we see that the proportional change in rotation speed in going from plane-polarized light to left- or right-circularly-polarized light is between 20% and 45%.

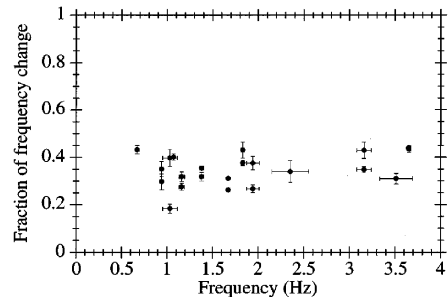


FIG. 5. Plot of fractional change in rotation frequency vs rotation frequency for CuO particles. The change in rotation frequency is due to rotation of a $\lambda/4$ plate by 45° , changing the polarization of the beam from plane polarized to circularly polarized. The error bars reflect the uncertainties in frequency measurements on the individual particles.

According to theory for the donut beam of charge 3, we expect to observe the torques on an absorbing particle trapped in a donut beam with left-circularly, plane-, and right-circularly-polarized light to be in the ratio 2:3:4 approximately. However, static friction between the particle and the slide can affect the ratio. As the particle size increases, we expect the torque required to initiate rotation to increase, causing all rotation frequencies to be reduced by approximately the same amount. This will have the effect of enlarging the proportional frequency change, which was indeed observed in our data.

In spite of this, the results show that, independent of frequency over a 4:1 range, the fractional frequency changes observed cluster about the expected 0.33 figure. Although the spread in values is too great to allow an accurate estimate of the correction term in Eq. (2) it can certainly be seen not to be dominant. Since, in practical terms, ours is a very strongly focused beam, this suggests that the simple paraxial

result that the total angular momentum of the field is a simple sum of orbital and spin components will be adequate in the great majority of cases.

The results also provide further proof that the rotation previously reported [5] is the result of optical angular-momentum transfer rather than, say, some thermal effect, as such effects certainly could not be polarization sensitive in this way.

Our experiments show that the rotational speed of small absorbing particles trapped in a focused laser beam depends on the polarization of the light. The dependence is consistent with the transfer of angular momentum to the particles from a field exhibiting two forms of angular momentum— an orbital component associated with the helical phase structure of the donut beam, and a spin component associated with circular polarization of the light— as expected for a paraxial Gauss-Laguerre mode.

-
- [1] R. Beth, *Phys. Rev.* **50**, 115 (1936).
 - [2] L. Allen, M.W. Beijersbergen, R.J.C. Spreeuw, and J.P. Woerdman, *Phys. Rev. A* **45**, 8185 (1992).
 - [3] G.T. di Francia, *Nuovo Cimento* **1**, 150 (1957).
 - [4] P. Allen, *Am. J. Phys.* **74**, 1185 (1966).
 - [5] H. He, M.E.J. Friese, N.R. Heckenberg, and H. Rubinsztein-Dunlop, *Phys. Rev. Lett.* **75**, 826 (1995).
 - [6] S.M. Barnett and L. Allen, *Opt. Commun.* **110**, 670 (1994).
 - [7] N. R. Heckenberg *et al.*, *Opt. Quantum Electron.* **24**, S951 (1992).
 - [8] H. He, N.R. Heckenberg, and H. Rubinsztein-Dunlop, *J. Mod. Opt.* **42**, 217 (1995).

Mechanical equivalence of spin and orbital angular momentum of light: an optical spanner

N. B. Simpson, K. Dholakia, L. Allen, and M. J. Padgett

J. F. Allen Physics Research Laboratories, Department of Physics and Astronomy, University of St. Andrews, North Haugh, St. Andrews, Fife KY16 9SS, Scotland

Received August 27, 1996

We use a Laguerre–Gaussian laser mode within an optical tweezers arrangement to demonstrate the transfer of the orbital angular momentum of a laser mode to a trapped particle. The particle is optically confined in three dimensions and can be made to rotate; thus the apparatus is an optical spanner. We show that the spin angular momentum of $\pm\hbar$ per photon associated with circularly polarized light can add to, or subtract from, the orbital angular momentum to give a total angular momentum. The observed cancellation of the spin and orbital angular momentum shows that, as predicted, a Laguerre–Gaussian mode with an azimuthal mode index $l = 1$ has a well-defined orbital angular momentum corresponding to $\hbar$ per photon. © 1997 Optical Society of America

The circularly symmetric Laguerre–Gaussian (LG) modes form a complete basis set for paraxial light beams.¹ Two indices identify a given mode, and the modes are normally denoted LG_p^l , where l is the number of 2π cycles in phase around the circumference and $(p + 1)$ is the number of radial nodes. In contrast to the planar wave fronts of the Hermite–Gaussian (HG) mode, for $l \neq 0$ the azimuthal phase term $\exp(il\phi)$ in the LG modes results in helical wave fronts.² These helical wave fronts are predicted by Allen *et al.* to give rise to an orbital angular momentum for linearly polarized light of $l\hbar$ per photon.³ This orbital angular momentum is distinct from the angular momentum associated with the photon spin manifested in circularly polarized light. For a collimated beam of circularly polarized light the photon spin is well known to be $\pm\hbar$; however, for a tightly focused beam the polarization state is no longer well defined.⁴ The total angular momentum of a beam, found by a rigorous solution of Maxwell's equations that reduces to a LG beam in the paraxial approximation, is given by⁵

$$\left[l + \sigma_z + \sigma_z \left(\frac{2kz_r}{2p + l + 1} + 1 \right)^{-1} \right] \hbar \quad (1)$$

per photon, where $\sigma_z = 0, \pm 1$ for linearly and circularly polarized light, respectively, and where z_r is the Rayleigh range and k is the wave number of the light. Note that for a collimated beam $kz_r \gg 1$, and expression (2) reduces to

$$(l + \sigma_z)\hbar \quad (2)$$

per photon, whereas for linearly polarized light $l\hbar$ per photon is strictly correct even in the absence of any approximation.

Lasers have been made that operate in LG modes,⁶ but it is easier to obtain them from the conversion of HG modes. Three different classes of mode converter have been demonstrated. Two of these, spiral phase plates⁷ and computer-generated holographic converters,^{8,9} introduce the azimuthal phase term to a $\text{HG}_{0,0}$ Gaussian beam to produce a LG mode with $p = 0$ and specific values of l . In all these devices a screw phase

dislocation, produced on axis, causes destructive interference, leading to the characteristic ring intensity pattern in the far field. The other class of converter is the cylindrical-lens mode converter,¹⁰ which employs the change in Gouy phase in the region of an elliptically focused beam to convert higher-order HG modes, of indices m and n , into the corresponding LG modes (or vice versa) with 100% efficiency. The transformation is such that $l = |m - n|$ and $p = \min(m, n)$.

Consider the various macroscopic mechanisms whereby angular momentum can be transferred between a well-defined transverse laser mode and matter. One can transfer spin angular momentum by using a birefringent optical component, resulting in a change in the polarization state of the light. One can transfer orbital angular momentum by using a component that possesses an azimuthal dependence of its optical thickness, as with a cylindrical lens or a spiral phase plate. In contrast, absorption of the light allows both spin and orbital angular momentum to be transferred.

The prediction that LG laser modes possess well-defined orbital angular momentum has led to considerable research activity.^{3,6–14} The transfer of orbital angular momentum from a light beam to small particles held at the focus has been modeled,¹² but to date only one group of researchers has demonstrated that these modes do indeed possess orbital angular momentum. He *et al.*¹³ demonstrated that orbital angular momentum could be transferred from the laser mode to micrometer-sized ceramic and metal-oxide particles near the focus of a LG beam with an azimuthal mode index $l = 3$. More recently the same authors reported the use of a circularly polarized mode and the corresponding speeding up or slowing down of the rotation, depending on the relative sign of the spin and orbital angular-momentum terms.¹⁴ In each case the particles were trapped with radiation pressure, first forcing the totally absorbing particle into the dark region on the beam axis (x – y trapping) and then forcing the particle against the microscope slide (z trapping). LG modes were recently also used in an optical tweezers

geometry for trapping hollow glass spheres¹⁵; however, no rotation was observed because these spheres were totally transparent.

Here we use weakly absorbing dielectric particles, larger than the dimensions of the tightly focused LG mode. The particle experiences a net force within the electric field gradient toward the focus of the beam and is thus held in an all-optical x - y - z trap. This clearly distinguishes our experiments from previous ones,^{13,14} which relied on mechanical restraint in the z direction. Such all-optical traps, using zero-order Gaussian modes, were demonstrated by Ashkin *et al.*¹⁶ and are commonly referred to as optical tweezers. The transfer of angular momentum and the subsequent rotation of the trapped particle through the use of a LG mode leads us to refer to our modified optical geometry as an optical spanner.

Figure 1 shows our experimental arrangement. We use a diode-pumped Nd:YLF laser at 1047 nm operating in a high-order HG mode. This mode is converted into the corresponding LG mode by the use of the previously mentioned cylindrical-lens mode converter; a quarter-wave plate sets the polarization state. A telescope arrangement optimizes the mode size to fill the back aperture of the objective lens, and an adjustable mirror enables the focused beam to be translated within the field of view of the objective lens. A dichroic beam splitter couples the beam into the optical spanner. The optical spanner is based on a 1.3-N.A., 100 $\times$ oil-immersion microscope objective. The associated tube lens forms an image of the trapped particle on a CCD array. The sample cell comprises a microscope slide, an ≈ 60 - μ m-thick vinyl spacer, and a cover slip and is backlit with a standard tungsten bulb. The LG mode is focused such that the trapped object is held in the focal plane of the objective lens, and a colored glass filter can be inserted before the CCD array to attenuate the reflected laser light. We independently confirmed the polarization state of the light in the plane of the trapped particle by monitoring the power transmitted through a linear polarizer at various orientations.

When the $l = 1$ LG mode was focused by the objective lens, its beam waist was approximately $0.8 \mu\text{m}$, which corresponds to a high-intensity ring $1.1 \mu\text{m}$ in diameter. For dielectric particles smaller than $\sim 1 \mu\text{m}$ we observe that the gradient force results in the particles' being trapped off axis, centered within the high-intensity region of the ring itself. However, particles $1 \mu\text{m}$ or larger are trapped on the beam axis and for incident power above a few milliwatts are held stably in three dimensions. If the particle is partially absorbing, the transfer of angular momentum from the laser beam to the particle causes the particle to rotate. The strong z -trapping force enables us to lift the particle optically off the button of the sample cell, allowing it to rotate more freely. We have successfully rotated a number of a different particle types, including particles of absorbing glass (e.g., Schott BG38 glass) and Teflon spheres. The particles were dispersed in various fluids, including water, methanol, and ethanol.

For the linearly polarized $l = 1$ LG mode and an incident power of ≈ 25 mW, typical rotation speeds for

the Teflon particles were ~ 1 Hz. The relationship between applied torque τ and limiting angular velocity ω_{lim} of a sphere of radius r in a viscous medium of viscosity η is¹⁷

$$\omega_{\text{lim}} = \tau / (8\pi\eta r^3). \quad (3)$$

This implies that the absorption of the Teflon particle is of the order of 2%.

The principal motivation of this study was to demonstrate that, with respect to absorption and the resulting mechanical rotation of a particle, the spin and the orbital angular momenta of light are equivalent. As discussed above, for the spin and orbital angular-momentum content of the laser mode to interact in an equivalent manner with the trapped particle it is essential that the dominant mechanism for transfer of the angular momentum be absorption. The birefringence or astigmatism of the trapped particle will preferentially rotate spin or orbital angular momentum, respectively. For comparison of spin and orbital angular momentum we selected Teflon particles, or amalgamated groups of Teflon particles, larger than the focused beam waist. Trapped on the beam axis, these particles interact uniformly with the whole of the beam, can be optically levitated off the bottom of the sample cell, and are observed to rotate smoothly with a constant angular velocity. This ability to lift the particles optically away from the cell boundary differentiates our research from that previously reported. Less regular particles could be lifted and made to rotate in the same sense as the more regular particles but often failed to stop when the spin and orbital angular-momentum terms were subtractive. This imperfect cancellation is due to additional transfer of orbital angular momentum owing to the asymmetry of the particle.

For our experimental configuration and an $l = 1$ mode, expression (1) shows that the total angular

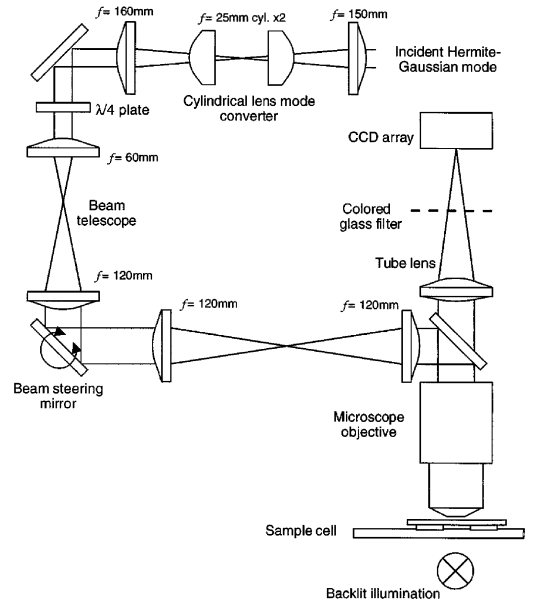


Fig. 1. Experimental configuration of the optical spanner.

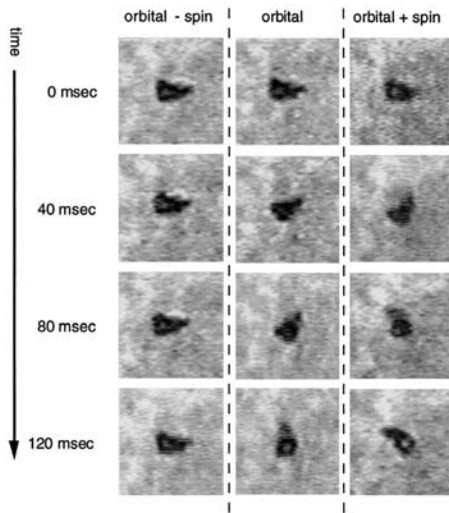


Fig. 2. Successive frames of the video image showing the stop-start behavior of a 2- μ m-diameter Teflon particle held with the optical spanner.

momentum is $\sim 2.06\hbar$ per photon when the spin and orbital terms are additive and $\sim 0.06\hbar$ per photon when the spin and orbital terms are subtractive.

Figure 2 shows successive frames of the video image of a 2- μ m-diameter Teflon particle held within the optical spanner, trapped with an $l = 1$ mode of various polarization states. The slight asymmetry in the particle geometry allows the particle rotation to be assessed. In more than 80% of cases, once a smoothly rotating particle had been selected, orienting the wave plate to circularly polarize the light would cause the particle to speed up or stop completely, depending on whether the spin and orbital angular-momentum terms were added or subtracted. In the other 20% of cases, although the particles would speed up or slow down, they could not be stopped completely. We believe that this was because these particles were insufficiently uniform, resulting in an unwanted mode transformation and giving an additional exchange of orbital angular momentum. With the well-behaved particles, when the spin and orbital terms are subtracted the low value of the torque, owing to a total angular momentum of no more than $0.06\hbar$ per absorbed photon, is insufficient to overcome the stiction present within the system, and the particle ceases to rotate. Stiction can arise from slight particle asymmetry coupled with residual astigmatism in the laser mode, which will favor the particles' being trapped in particular orientations. When the spin and orbital terms are additive the rotation speed increases significantly, but in only few cases is it seen to double. We attribute this to a nonlinear relationship between the applied torque and the terminal rotation speed, and the particle asymmetry. We confirmed this nonlinearity in the relationship between torque and rotation speed by deliberately changing the optical power while maintaining the same laser mode.

We observe, in agreement with our interpretation, that reversing the sense of the cylindrical-lens mode converter caused the particles to rotate in the opposite direction; similarly, by use of a $HG_{0,0}$ mode, particles could be rotated in either direction when circularly polarized light of the appropriate handedness was used.

Our experiment uses a LG mode within an all-optical x - y - z trap to form an optical spanner for rotating micrometer-sized particles. By controlling the polarization state of a LG mode with $l = 1$ we can arrange for the angular momentum associated with the photon spin to add to the orbital angular momentum, giving a total angular momentum of $2\hbar$ per photon, in which case the Teflon particle spins more quickly. Alternatively, that momentum can subtract, resulting in no overall angular momentum, and the particle comes to a halt. We observe the cancellation of the spin and orbital angular momentum in a macroscopic system, which verifies that for an $l = 1$ LG mode the orbital angular momentum is indeed well defined and corresponds to $\hbar$ per photon.

This research is supported by Engineering and Physical Sciences Research Council grant GR/K11536. M. I. Padgett is a Royal Society Research Fellow.

References

1. A. E. Siegman, *Lasers* (University Science, Mill Valley, Calif., 1986), Sec. 17.5, pp. 685–695.
2. J. M. Vaughan and D. V. Willets, *Opt. Commun.* **30**, 263 (1979).
3. L. Allen, M. W. Beijersbergen, R. J. C. Spreeuw, and J. P. Woerdman, *Phys. Rev. A* **45**, 8185 (1992).
4. D. N. Pattanayak and G. P. Agrawal, *Phys. Rev. A* **22**, 1159 (1980).
5. S. M. Barnett and L. Allen, *Opt. Commun.* **110**, 670 (1994).
6. M. Harris, C. A. Hill, and J. M. Vaughan, *Opt. Commun.* **106**, 161 (1994).
7. M. W. Beijersbergen, R. P. C. Coerwinkel, M. Kristensen, and J. P. Woerdman, *Opt. Commun.* **112**, 321 (1994).
8. N. R. Heckenberg, R. McDuff, C. P. Smith, and A. G. White, *Opt. Lett.* **17**, 221 (1992).
9. N. R. Heckenberg, R. McDuff, C. P. Smith, H. Rubinsztein-Dunlop, and M. J. Wegener, *Opt. Quantum Electron.* **24**, S951 (1992).
10. M. W. Beijersbergen, L. Allen, H. E. L. O. van der Veen, and J. P. Woerdman, *Opt. Commun.* **96**, 123 (1993).
11. M. Babiker, W. L. Power, and L. Allen, *Phys. Rev. Lett.* **73**, 1239 (1994); S. J. van Enk and G. Nienhuis, *Opt. Commun.* **94**, 147 (1992).
12. N. B. Simpson, L. Allen, and M. J. Padgett, *J. Mod. Opt.* **43**, 2485 (1996).
13. H. He, M. E. J. Friese, N. R. Heckenberg, and H. Rubinsztein-Dunlop, *Phys. Rev. Lett.* **75**, 826 (1995).
14. M. E. J. Friese, J. Enger, H. Rubinsztein-Dunlop, and N. R. Heckenberg, *Phys. Rev. A* **54**, 1593 (1996).
15. K. T. Gahagan and G. A. Swartzlander, Jr., *Opt. Lett.* **21**, 827 (1996).
16. A. Ashkin, J. M. Dziedzic, J. E. Bjorkholm, and S. Chu, *Opt. Lett.* **11**, 288 (1986).
17. S. Oka, in *Rheology*, F. R. Eirich, ed. (Academic, New York, 1960), Vol. 3.

